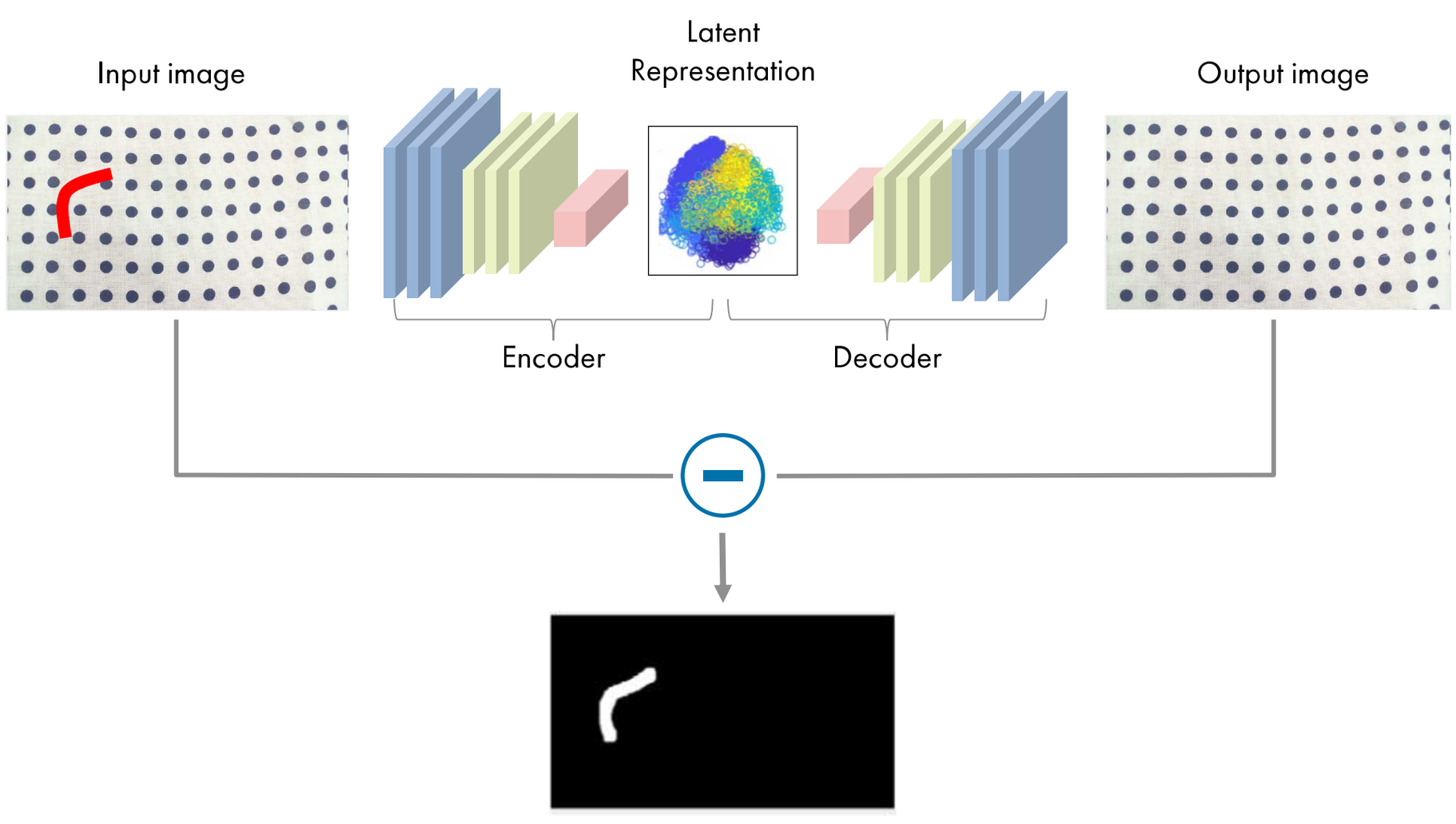
Lecture 1b

Variational Autoencoders (VAE): from Autoencoders to Generative Models

What is an Autoencoder?

An Autoencoder (AE) is a type of artificial neural network used primarily for learning efficient data representations (encoding) and reconstructing the input from this representation (decoding).



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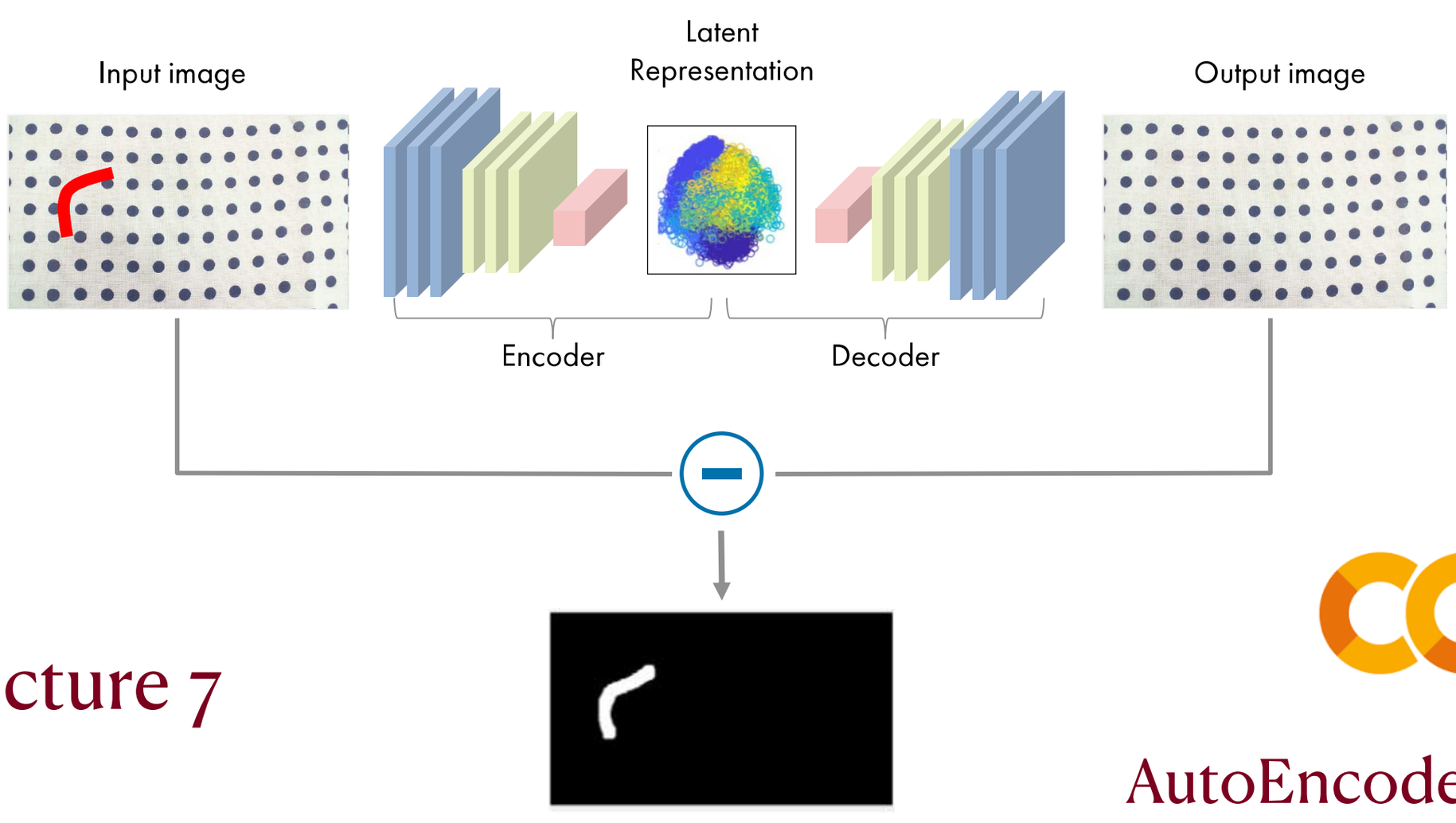
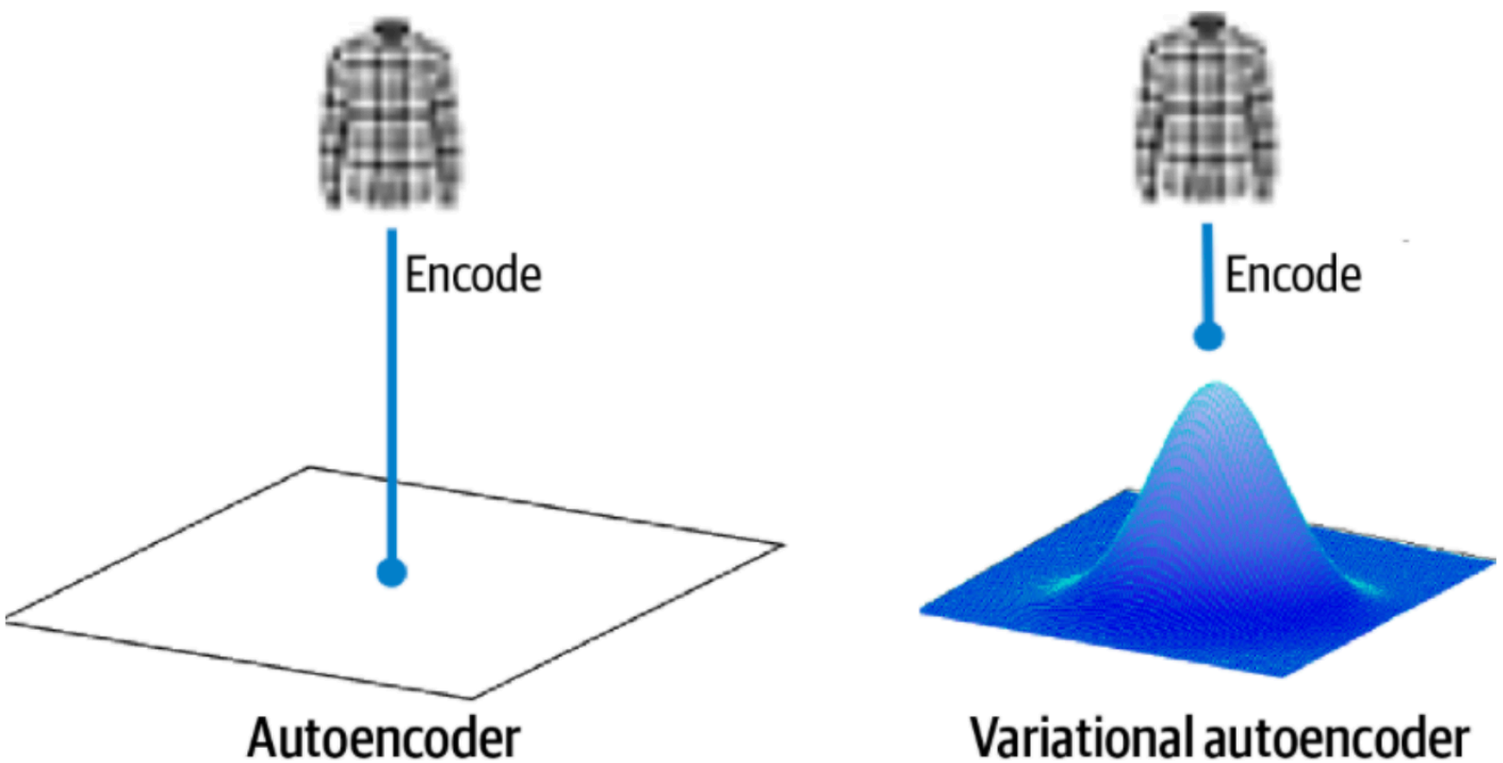


Lecture 7

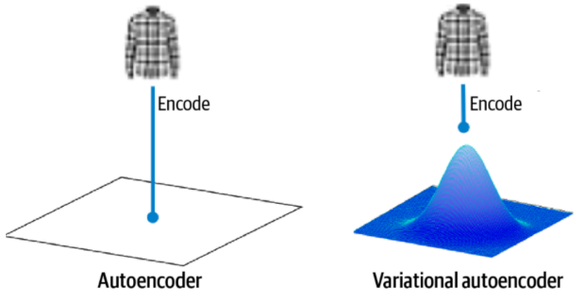
AutoEncoder_3.ipynb

Limitations of Autoencoders

- **No Probabilistic Interpretation:**
  - The latent space is **deterministic**, meaning each input is mapped to a single latent vector, not a distribution.
- **Unstructured Latent Space:**
  - Latent space lacks continuity or smooth structure, making interpolation or sampling difficult or unreliable.
- **Limited Generative Capability:**
  - Autoencoders cannot easily generate new, realistic samples by randomly sampling the latent space.
- **Risk of Overfitting:**
  - Potentially memorizing the data rather than learning generalizable latent representations, limiting generative or exploratory analysis.



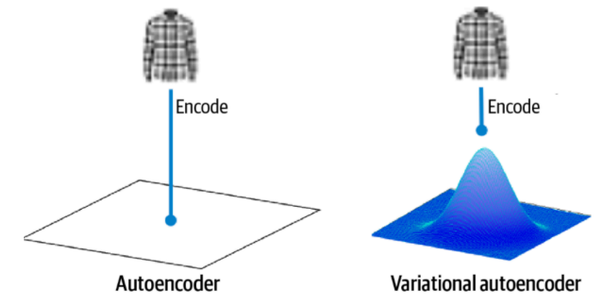
Motivation for Variational Autoencoders



Traditional Autoencoders face several critical limitations, motivating the development of **Variational Autoencoders (VAEs)**:

- **Structured Latent Spaces:**
  - **Autoencoders** often produce **irregular** and discontinuous latent spaces.
  - **Variational Autoencoders** encourage structured and continuous latent representations.
- **Generative Capabilities:**
  - VAEs explicitly define a **probability distribution** over latent space, enabling the generation of new data by sampling.
- **Probabilistic Interpretation:**
  - Allow quantifying **uncertainty** and variability, leading to more robust and interpretable latent features.
- **Smooth Interpolation:**
  - Smooth transitions between points in latent space facilitate meaningful data **interpolation and exploration**.

Introduction to Variational Autoencoders (VAE)



- **Definition:**

  > A **Variational Autoencoder (VAE)** is a generative model that extends the Autoencoder framework by introducing a probabilistic latent space. It encodes data into a probability distribution, allowing for the generation of new data samples.

- **Key Idea:**
  - **Encoder** maps inputs not to fixed points, but to a probability distribution (usually Gaussian) in latent space.
  - **Decoder** generates new, plausible data points by sampling from this latent distribution.

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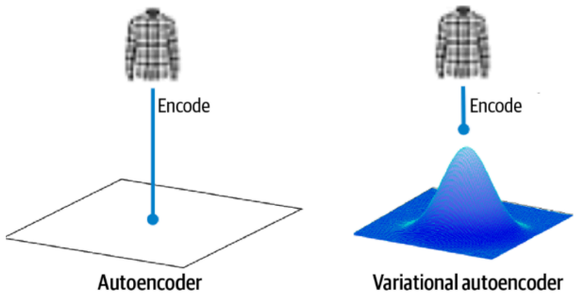

Main Differences with Autoencoders:

Aspect	Autoencoder (AE)	Variational Autoencoder (VAE)
Latent Representation	Single vector (deterministic)	Probability distribution (probabilistic)
Generative Ability	Limited	Strong generative capabilities
Training	Reconstruction loss only	Reconstruction + KL divergence

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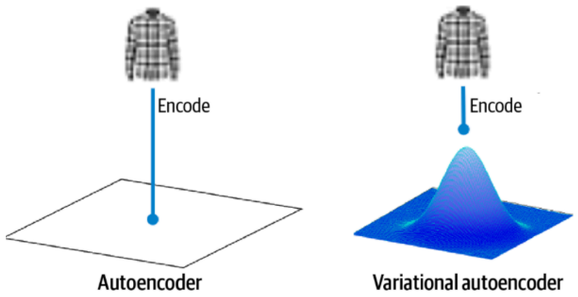

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Loss Function (Reconstruction Loss):

- Typically Mean Squared Error (MSE):

$$\mathcal{L}(x, \hat{x}) = \|x - \hat{x}\|^2 = \|x - g_\phi(f_\theta(x))\|^2$$

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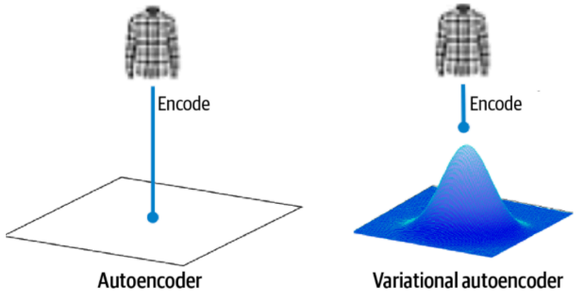

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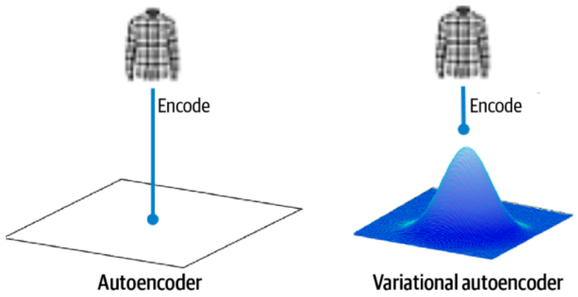

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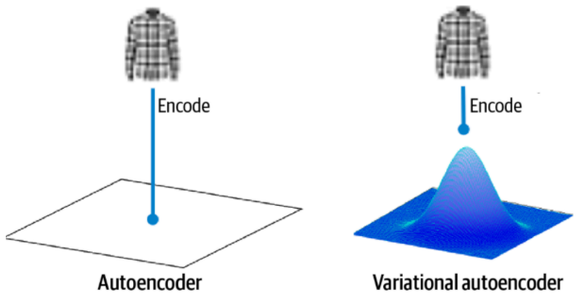

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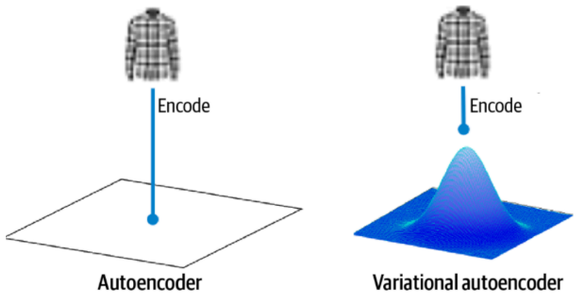

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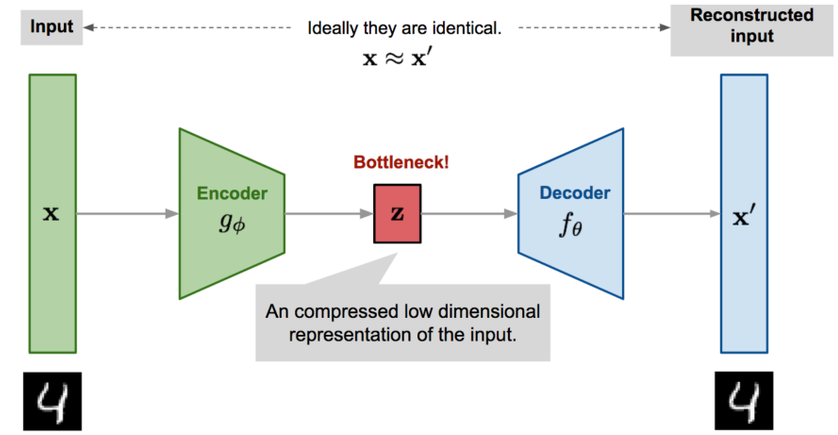
- **Kingma & Welling (2014):**
  - "Auto-Encoding Variational Bayes" (arXiv:1312.6114)

Architecture of Variational Autoencoder



1. Encoder (Inference Model):

- Maps input data to a **distribution in latent space**, characterized by parameters:

$$\mu(x), \quad \sigma^2(x)$$

- Produces a latent variable distribution:

$$z \sim \quad q_\phi(z|x) = \mathcal{N}(z; \mu_\phi(x), \sigma_\phi^2(x))$$

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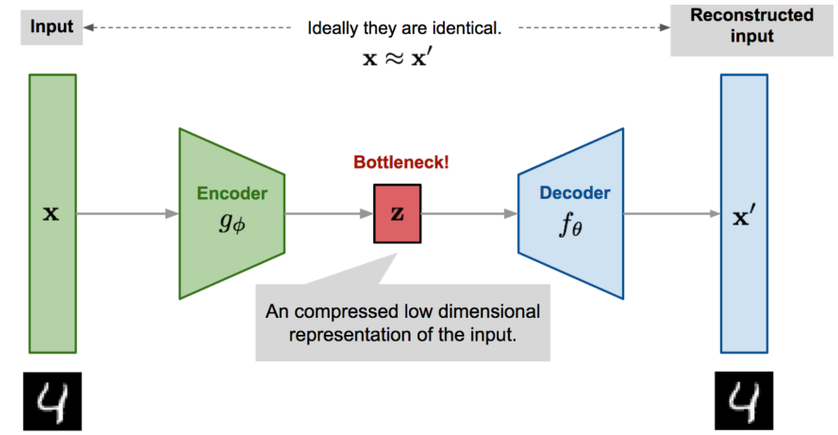

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Diagonal Covariance Matrix=Vector

(Independent features in latent space)

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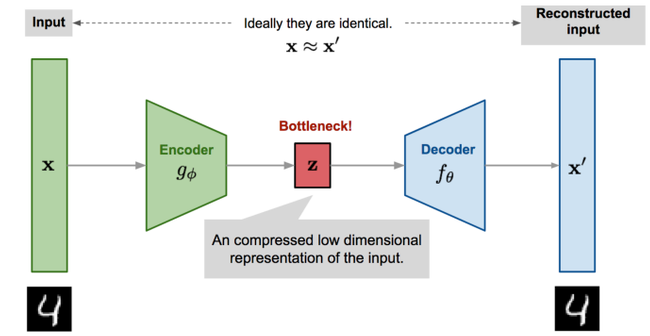

Decoder (Generative Model)

- Given a sampled latent variable z, the decoder reconstructs the data x using parameters θ:

$$\hat{x} \sim p_\theta(x|z) = \mathcal{N}(x;\, \mu_\theta(z),\, \sigma_\theta^2(z))$$

Parameters Explanation:

- **Mean Reconstruction**: $\mu_\theta(z)$
  - The decoder generates the expected reconstructed data from latent representation z.
- **Reconstruction Variance**: $\sigma_\theta^2(z)$
  - Often fixed (e.g., to a constant) or ignored in basic VAE implementations, especially for simplicity.

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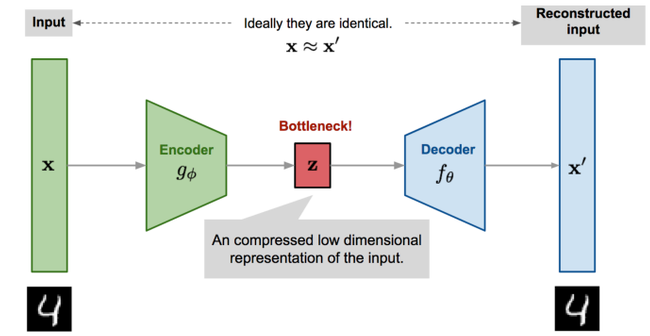

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Reparameterization Trick:

- To enable efficient gradient descent, stochastic sampling is reparameterized:

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

$$q_\phi(z|x) = \mathcal{N}(z; \mu_\phi(x), \sigma_\phi^2(x))$$

A step Back (2014)

Auto-Encoding Variational Bayes

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Abstract

How can we perform efficient inference and learning in directed probabilistic models, in the presence of continuous latent variables with intractable posterior distributions, and large datasets? We introduce a stochastic variational inference and learning algorithm that scales to large datasets and, under some mild differentiability conditions, even works in the intractable case. Our contributions are two-fold. First, we show that a reparameterization of the variational lower bound yields a lower bound estimator that can be straightforwardly optimized using standard stochastic gradient methods. Second, we show that for i.i.d. datasets with continuous latent variables per datapoint, posterior inference can be made especially efficient by fitting an approximate inference model (also called a recognition model) to the intractable posterior using the proposed lower bound estimator. Theoretical advantages are reflected in experimental results.

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- **Likelihood:**
  - Describes how likely the data is, given a latent variable.
  - Formally: $p_\theta(x|z)$.
  - Estimated by the decoder.

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Example: Recognizing handwritten digits (MNIST)

- **Prior**: You initially assume each latent variable z has a certain simple distribution (no observation yet).
- **Posterior**: After seeing the digit image, your model updates the belief, producing a specific latent distribution centered on this digit's representation.
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Bayes' Theorem (General form):

$$p(z|x) = \frac{p(x|z)\,p(z)}{p(x)}$$

$$p(\boldsymbol{x}) = \int p(\boldsymbol{x}, \boldsymbol{z}) d\boldsymbol{z}$$

$$p(x) = \int p(x|z)p(z)\,dz$$

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Encoder

Posterior $$p(z|x) = \frac{p(x|z)\,p(z)}{p(x)}$$

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Intuitive Explanation (in the context of VAE):

- **Prior**: Your initial belief about latent structure (usually a simple Gaussian distribution).
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Why this matters for VAEs:

- Computing exact posterior $p(z|x)$ is intractable.
- **Variational Autoencoders** use an approximate posterior $q_\phi(z|x)$ to solve this challenge efficiently.
- This approximation simplifies the problem into optimizing the ELBO, balancing reconstruction accuracy (likelihood) and regularization (keeping posterior close to the prior).

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Relationship between Sampling $z \sim p(z)$ and Distribution $p(x)$:

- When you generate data points by first sampling latent vectors z from the prior distribution $p(z)$ and then using the decoder distribution $p_\theta(x|z)$:

$$z \sim p(z), \quad x \sim p_\theta(x|z)$$

- **These generated samples** x indeed come from the model's implicit marginal distribution:

$$p_\theta(x) = \int p_\theta(x|z)p(z)\,dz$$

- In other words, **by sampling from the prior over** z and then applying the decoder, you're effectively sampling from your learned model's marginal distribution $p_\theta(x)$.

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- Instead of computing $p(x)$ exactly, VAEs use a lower bound (**ELBO**) to approximate and optimize it:

$$\log p(x) \geq \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - KL(q_\phi(z|x)\|p(z))}_{\text{ELBO}}$$

- The VAE doesn't need the exact value of $p(x)$, only the relative ELBO to guide training.
- Maximizing ELBO indirectly increases the likelihood of the data $p(x)$.

Step-by-step: Training a Variational Autoencoder

Step 1: Setup and Goal

- **Goal of Training:**
  - Find encoder (ϕ) and decoder (θ) parameters to maximize likelihood of observed data x:

$$\text{Maximize} \quad \log p_\theta(x)$$

- **Issue:**
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Why do we maximize $\log p(x)$ if we ultimately want to reproduce $p(x)$?

1. The Goal of Generative Models (like VAE):

- Our ultimate goal is indeed **to reproduce (or approximate) the true data distribution** $p_{\text{data}}(x)$.
- This means we want our **model distribution**, which we call $p_\theta(x)$, to closely match the true data distribution $p_{\text{data}}(x)$.

Mathematically, our aim is:

$$p_\theta(x) \approx p_{\text{data}}(x)$$

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2. How do we measure the closeness between distributions?

- One common and mathematically convenient measure to quantify "closeness" is the **Kullback–Leibler divergence** (KL divergence), which measures how one distribution differs from another:

$$\mathrm{KL}(p_{\text{data}}(x) \| p_\theta(x)) = \int p_{\text{data}}(x) \log \frac{p_{\text{data}}(x)}{p_\theta(x)} \, dx$$

- Minimizing this KL divergence means minimizing the difference between the data distribution and our model's learned distribution.

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3. Connection to maximizing $\log p(x)$:

- Notice that the KL divergence can be written as:

$$KL(p_{\text{data}}(x) \| p_\theta(x)) = \mathbb{E}_{p_{\text{data}}(x)}[\log p_{\text{data}}(x) - \log p_\theta(x)]$$

- The term $\log p_{\text{data}}(x)$ is **fixed** and doesn't depend on our model parameters (θ). Thus, minimizing KL divergence is equivalent to:

$$\text{Minimize } KL(p_{\text{data}}(x) || p_\theta(x)) \quad \Longleftrightarrow \quad \text{Maximize } \mathbb{E}_{p_{\text{data}}(x)}[\log p_\theta(x)]$$

This expectation $\mathbb{E}_{x \sim p_{\text{data}}}[\log p_\theta(x)]$ is exactly what we call the **log-likelihood**.

Step-by-step: Training a Variational Autoencoder

Step 1: Setup and Goal

- **Goal of Training:**
  - Find encoder (ϕ) and decoder (θ) parameters to maximize likelihood of observed data x:

$$\text{Maximize} \quad \log p_\theta(x)$$

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Step 2: Introducing the Evidence Lower Bound (ELBO)

- Instead of maximizing $\log p(x)$ directly, VAEs maximize a **lower bound** called ELBO:

$$\log p(x) \geq \mathcal{L}_{\phi,\theta}(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - KL(q_\phi(z|x)\|p(z))$$

- Maximizing ELBO indirectly maximizes the data likelihood.

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Step 3: Interpreting ELBO Components

ELBO has two clear components:

- **Reconstruction Term** (Decoder's accuracy):

$$\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$$

- **KL Divergence** (Regularization):

$$-\mathrm{KL}(q_\phi(z|x)\|p(z))$$

- **Intuition**:
  - Reconstruction encourages accurate generation of data.
  - KL divergence keeps latent distributions $q_\phi(z|x)$ close to the prior $p(z)$.

Why the reconstruction term measures decoder accuracy:

Recall the reconstruction term mathematically defined as:

$$\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$$

To see intuitively why this term measures accuracy in reconstruction, we proceed as follows:

Why the reconstruction term measures decoder accuracy:

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To see intuitively why this term measures accuracy in reconstruction, we proceed as follows:

Step 1: Intuition (informal):

- The **decoder** tries to **reproduce input** x given latent variables z.
- The decoder defines a conditional probability distribution $p_\theta(x|z)$, which measures **how probable or likely the original data** x **is, given the latent representation** z.

In other words:

- High $p_\theta(x|z)$ means: "Given latent variables z, the decoder produces data very similar to the original input x."
- Low $p_\theta(x|z)$ means the decoder reconstructs poorly from the latent variables z.

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Step 2: Why the expectation $\mathbb{E}_{q_\phi(z|x)}$?

- The latent variable z is not deterministic but sampled from a distribution $q_\phi(z|x)$, produced by the encoder.
- To evaluate decoder accuracy fairly, you measure its performance on average across multiple possible latent representations produced by the encoder.
- This means you assess reconstruction accuracy averaged over all latent variables that the encoder could produce given input x.

Why the reconstruction term measures decoder accuracy:

Recall the reconstruction term mathematically defined as:

$$\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$$

To see intuitively why this term measures accuracy in reconstruction, we proceed as follows:

"On average, how well does my decoder reconstruct the input x, if I use latent representations z sampled from my encoder's approximate posterior?"

Why the reconstruction term measures decoder accuracy:

Recall the reconstruction term mathematically defined as:

$$\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$$

To see intuitively why this term measures accuracy in reconstruction, we proceed as follows:

In practice, the decoder often assumes a Gaussian likelihood (for continuous data, like images or signals):

$$p_\theta(x|z) = \mathcal{N}(x; \mu_\theta(z), \sigma^2 I)$$

Then, the log-likelihood becomes:

$$\log p_\theta(x|z) \propto -||x - \mu_\theta(z)||^2$$

Here, clearly, maximizing this log-likelihood is the same as minimizing the reconstruction error, making the decoder's output $\mu_\theta(z)$ closer and closer to the actual input x.

- Instead of computing $p(x)$ exactly, VAEs use a lower bound (**ELBO**) to approximate and optimize it:

$$\log p(x) \geq \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - KL(q_\phi(z|x)\|p(z))}_{\text{ELBO}}$$

- The VAE doesn't need the exact value of $p(x)$, only the relative ELBO to guide training.
- Maximizing ELBO indirectly increases the likelihood of the data $p(x)$.

2. KL Divergence (Regularization Term):

$$-\mathrm{KL}(q_\phi(z|x)\|p(z))$$

- **Meaning:**
  - Measures how "far" the approximate posterior $q_\phi(z|x)$ (encoder's distribution) deviates from the prior distribution $p(z)$, usually $\mathcal{N}(0, I)$.
- **Practical interpretation:**
  - Ensures that latent representations stay structured, continuous, and compact.
  - Prevents the encoder from learning overly specific or "isolated" encodings.
- **Mathematical intuition:**
  - $\mathrm{KL}(q_\phi(z|x)\|p(z)) = 0$ when encoder's distribution exactly matches prior.
  - Increasing this term (minimizing divergence) improves generalization and enables latent space smoothness, facilitating meaningful data generation.
- **Explicit formula (assuming Gaussian distributions)**:

$$KL\left(\mathcal{N}(\mu_\phi(x), \sigma_\phi(x)) \,||\, \mathcal{N}(0, I)\right) = \frac{1}{2}\sum_j \left[\mu_{\phi,j}^2(x) + \sigma_\phi^2(x) - 1 - \log(\sigma_\phi^2(x))\right]$$

Now ‘VAE_4.ipynb’

Then we go to a more evolved 'convolutional' VAE on a bigger data set (CELEBA): see Lecture 8b on Convolution and “VAE_Convolution_CELEBA_5.ipynb".....then VQ_VAE...